EE3-03

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

EEE/ISE PART III/IV: MEng, BEng and ACGI

Corrected Copy

# COMMUNICATION SYSTEMS

Tuesday 9 December 2014, 9:00 am

Time allowed: 3:00 hours

**There are 23 questions on this paper.**

**Answer ALL questions.**

**The multiple choice questions together account for 40% of the marks.**

*Answers to multiple choice questions 1-10 should be given on the paper itself.*

*Students are not permitted to use more than one answer book.*

*Students are not permitted to take the question paper away.*

*The following are provided:*
*A table of Fourier transforms*
*A Gaussian Tail Function graph*

Examiners responsible:

First Marker(s): A. Manikas
Second Marker(s): B. Clerckx

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## PART-I

1. Consider two analogue signals x($t$) and y($t$). The signal x($t$) has a uniform amplitude probability density function, i.e.

$$\text{pdf}_x(x) = \frac{1}{3}\text{rect}\left\{\frac{x}{3}\right\}$$

while the signal y(t) has a Gaussian amplitude probability density function $\text{pdf}_y(y)$ of mean $\mu_y$ and standard deviation equal to $\sqrt{P_x}$where $P_x$ denotes the average power of x($t$).

Which of the following is correct? [4 marks]

(a) the average power $P_x$ of the signal x($t$) is $\frac{1}{9}$,

(b) the differential entropy H(x) of x($t$) is 1.8396,

(c) the differential entropy H(y) of y($t$) is 1.8396,

(d) the entropy power of x(t) is 0.52698,

(e) the entropy power of y(t) is 0.52698.

2. For a speech signal transmitted using a uniform quantiser of 256 levels the bit rate at the output of the source encoder is [4 marks]

(a) 8 kbits/s,

(b) 16 kbits/s,

(c) 32 kbits/s,

(d) 64 kbits/s,

(e) 128 kbits/s.

3. A digital communication system having an energy utilisation efficiency (EUE) equal to 30, operates in the presence of additive white Gaussian noise of double-sided power spectral density $\text{PSD}_n(f)$=$0.5\times10^{-6}$ W/Hz. If the channel capacity $C$ is 16 kbits/s and the data bit rate $r_b$ is 2 kbits/s, then the channel bandwidth $B$ is [4 marks]

(a) 4 kHz,

(b) 8 kHz,

(c) 16 kHz,

(d) 32 kHz,

(e) 64 kHz.

4. An analogue message signal $g(t)$ of maximum frequency $F_g$ is sampled at the Nyquist rate and then quantized to $Q = 16$ levels. A source encoder encodes the output with $\log_2 Q$ bits per level. Then a digital modulator transmits each binary digit through a bandlimited channel, as one of two energy signals in the presence of white Gaussian noise of double-sided power spectral density $10^{-5}$ Watts/Hz. The energy $E_b$ necessary to satisfy the condition $\mathrm{SNR}_{in-mb} = 30\mathrm{dB}$ is [4 marks]

   (a) $3 \times 10^{-3}$,
   (b) $2.5 \times 10^{-3}$,
   (c) $2 \times 10^{-3}$,
   (d) $1.5 \times 10^{-3}$,
   (e) none of the above.

5. Consider the bipolar line code waveform $\mathrm{b(t)} = \sum_{n=-\infty}^{\infty} \mathrm{a}[n].c(t - nT_{cs})$ where the autocorrelation function of the code sequence $\{\mathrm{a}(n)\}$ is as follows:

$$R_{\mathrm{aa}}(k) = \begin{cases} 1/2 & \text{if } k = 0 \\ -1/4 & \text{if } k = 1 \\ 0 & \text{if } k \geq 2 \end{cases}$$

   If $\Pr(0) = \Pr(\pm 1) = 0.5$ and $c(t) = \mathrm{rect}\left\{\frac{t}{T_{cs}}\right\}$ then the power spectral density of b(t) is [4 marks]

   (a) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}$,
   (b) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}$,
   (c) $T_{cs}.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$,
   (d) $T_{cs}^2.\mathrm{sinc}^2\{fT_{cs}\}.\sin^2(\pi T_{cs})$,
   (e) none of the above.

6. A FH/SSS has a data rate of 4800 bits/second and uses 4 hops per data bit. What is the bandwidth if there are 1024 hopping frequencies? [4 marks]

   (a) 19.2 kHz.
   (b) 4.096 MHz.
   (c) 4.915 MHz.
   (d) 19.661 MHz.
   (e) None of the above.

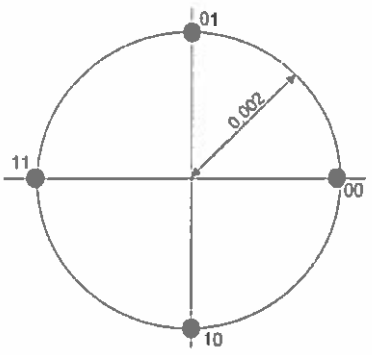
7. The minimum chip rate needed by a DS-BPSK CDMA system to resolve a multipath, with an additional path length of 15m compared to the direct path, is [4 marks]

   (a) 10 Mchips/sec,
   (b) 20 Mchips/sec,
   (c) 40 Mchips/sec,
   (d) 60 Mchips/sec,
   (e) none of the above.

8. Consider a CDMA system of 256 users where each user has a protection probability equal to $4 \times 10^{-2}$ and an Anti-jam Margin of 30 dB. Each user employs a feedback shift register of 21 stages, whose feedback connections are described by a primitive polynomial. The system is perfectly power controlled and the received power from each user is equal to $P = 0.1915$W operating in the presence of additive white Gaussian noise of double sided power spectral density $0.5 \times 10^{-6}$ Watts/Hz. The average energy per bit $E_b$ is [4 marks]

   (a) $0.94 \times 10^{-3}$,
   (b) $1.881 \times 10^{-3}$,
   (c) $2.33 \times 10^{-3}$,
   (d) $4.46 \times 10^{-3}$,
   (e) none of the above.

9. The constellation diagram of a QPSK Communication system is shown below




   If the bit rate is 40kbits/sec what is the amplitude of this $M$-ary set of signals? [4 marks]

   (a) $\sqrt{8}$mV.
   (b) 0.16V.
   (c) 0.4 V.
   (d) 0.8 V.
   (e) None of the above.

10. A speech signal having a maximum frequency of 4kHz is sampled at twice the Nyquist rate and then fed through an 8-bit uniform quantizer. The generated binary sequence is then fed through a binary PSK direct sequence spread spectrum system which operates in the presence of a jammer of power 1.6 Watts and in the presence of additive white Gaussian noise with double-sided power spectral density $0.5 \times 10^{-12}$ Watts/Hz. The amplitude of the BPSK signal is 0.5V.

    For this system, the spread spectrum bandwidth $B_{ss}$ is 32 MHz and the system is not synchronised.

    The power of the jammer at the output of the despreader is [4 marks]

    (a) 1.6 $m$W,

    (b) 3.2 $m$W,

    (c) 6.4 $m$W,

    (d) 12.8 $m$W,

    (e) none of the above.

## PART-II

11. A binary digital communication system, having an energy utilisation efficiency (EUE) equal to 40, operates in the presence of additive white Gaussian noise of double-sided power spectral density PSD($f$)=0.5106 W/Hz. If the channel capacity $C$ is 16 kbits/s and the channel bandwidth $B$ is 4 kHz, estimate

    (a) the noise power at the channel output; [1 marks]

    (b) the bit rate $r_b$. [3 marks]

12. Consider a random binary sequence of 0's and 1's, which are statistically independent with $\Pr(1) = \Pr(0) = 0.5$. This binary sequence is transmitted as a random signal with 1's and 0's being sent using the pulses $s_1(t)$ and $s_0(t)$ described below:

$$0 \longmapsto s_0(t) = -3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

    and

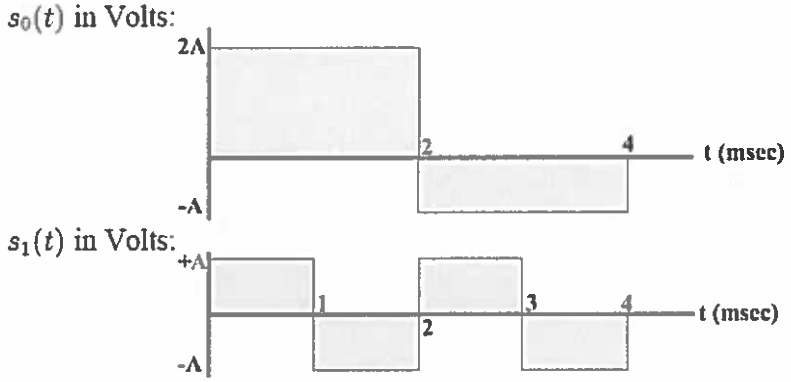
$$1 \longmapsto s_1(t) = +3 \times 10^{-3} \times \Lambda\left\{\frac{t}{1ms}\right\} \text{ Volts}$$

    (a) Find the Power Spectral Density of the transmitted signal. [4 marks]

    (b) What is the dimensionality $D$ of the signal space? [1 mark]

    (c) Plot the constellation diagram. [2 marks]

13. Given a binary digital communication system which employs the following two equally-probable signals and an optimum digital demodulator (matched filter)




    The received signals are corrupted by additive white Gaussian channel noise having a double-sided power spectral density of $\frac{7}{8}10^{-9}$ W/Hz.

    If the amplitude A is equal to 1mV, find the bit-error rate. [4 marks]

14. Encode, using HDB3 code, the following two binary sequences [4 marks]

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |

15. Consider a feedback shift register whose feedback connections are represented by the primitive polynomial $D^5 + D^2 + 1$ and operates with a clock rate 2.7Mb/sec.

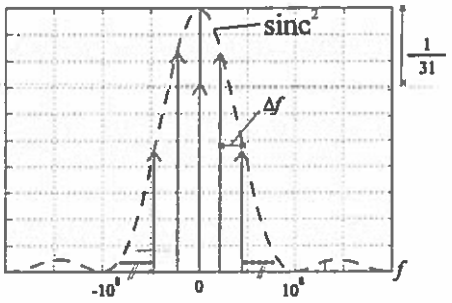
    (a) Sketch this feedback shift register [2 marks]

    (b) Find the period of the output sequence in minutes. [1 mark]

    (c) Give the first 7 bits of one period of its output sequence - starting with all 1's (initial condition). [2 marks]

16. A pseudo random (PN) signal $b(t)$ is generated by using a maximal length shift register of $m$-stages and has the following double-sided Power Spectral Density.

    $\text{PSD}_b(f)$ :




    (a) Find the number $m$ of shift register stages. [2 marks]

    (b) Find $\Delta f$. [2 marks]

17. Consider a PCM system where its sampler operates at twice the Nyquist rate and its quantizer consists of a $\mu$-law compander (with $\mu = 100$) followed by a $Q$-level uniform quantizer.
    Note that $\mu$-law compression is defined as follows:

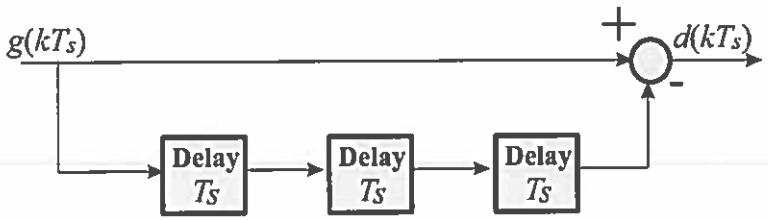
$$\text{output} = \frac{\ln(1+\mu.\,|\,x\,|)}{\ln(1+\mu)}\text{sign}(x) \qquad \text{where } x = \frac{\text{input value in } Volts}{\text{maximum input value in } Volts}$$

    The maximum and minimum values of the input signal are +5Volts and −5Volts respectively and the signal occupies a bandwidth from 100Hz to 15kHz. It is required that the Signal-to-Quantization-Noise Ratio (SNRq) should be better than 8.5 dB

    (a) Find the minimum number of quantizing levels $Q$ required, as well as the transmission bit rate. [2 marks]

    (b) If the signal at the output of the sampler at time $kT_s$ is equal to $-2.4$ Volts, estimate the instantanous quantization noise $n_q(kT_s)$ where $T_s$ is the sampling period and $k$ is an integer. [4 marks]

18. Consider a non-white zero-mean Gaussian random source $g(t)$ having an autocorrelation function $R_{gg}(\tau) = \exp(-8000 \mid \tau \mid)$. The signal is sampled at a rate of 24 ksamples/sec and then is applied at the input the following circuit




For the signal $d(kT_s)$, where $T_s$ is the sample time, $k$ is an integer, find:

(a) the $DC$ level, [1 mark]

(b) the average power. [3 marks]

19. A high quality music signal with a Crest Factor of 4.4668, having a maximum frequency 18 kHz, is applied to a uniform PCM system (i.e. PCM with a uniform quantizer). If it is specified that the Signal-to-Quantisation-Noise ratio ($SNR_q$) should be better than 50dB, find the minimum data bit rate required [6 marks]

20. In a binary PCM communication system prove that bandwidth expansion factor $\beta$ is equal to the average number of bits $\gamma$ per quantization level, i.e. [4 marks]

$$\beta = \gamma$$

21. The two signals $s_0(t)$ and $s_1(t)$ of a binary communication system each have energy equal to 93.3 and cross correlation coefficient 0.866.

(a) Draw the constellation diagram of the system properly labeled. [1 mark]

(b) What is the distance of these two signals? [3 marks]

22. Define the terms in a CDMA wireless channel:

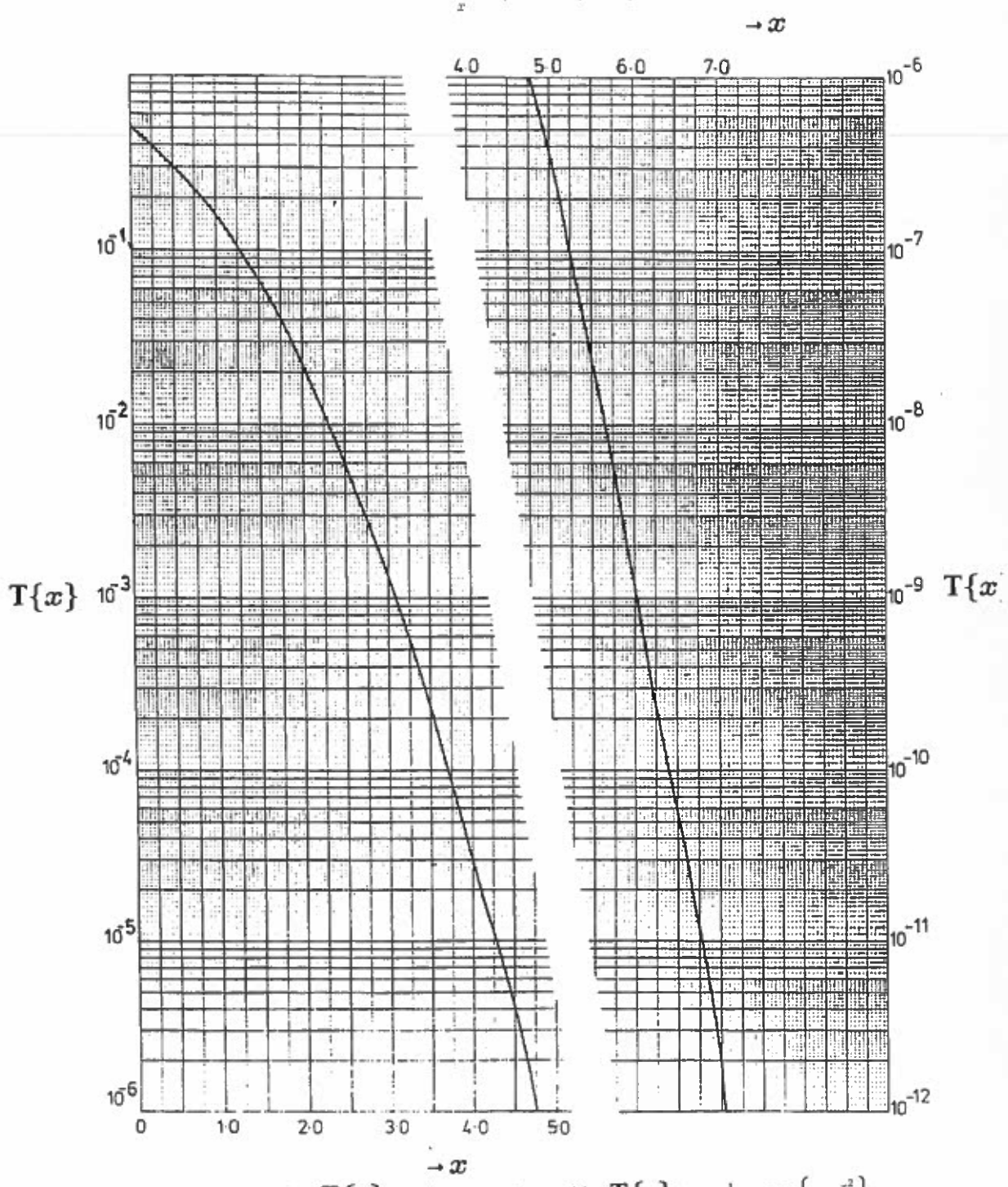
- delay spread, coherence bandwidth and flat fading . [3 marks]
- channel selectivity and channel coherence [2 marks]

23. Describe the log-distance path loss model of a wireless channel. [3 marks]

| Fourier Transform Tables | | | |
|---|---|---|---|
| | Description | Function | Transformation |
| 1 | Definition | $g(t)$ | $G(f) = \int_{-\infty}^{\infty} g(t).e^{-j2\pi ft}dt$ |
| 2 | Scaling | $g\left(\frac{t}{T}\right)$ | $\lvert T\rvert .G(fT)$ |
| 3 | Time shift | $g(t-T)$ | $G(f).e^{-j2\pi fT}$ |
| 4 | Frequency shift | $g(t).e^{j2\pi Ft}$ | $G(f-F)$ |
| 5 | Complex conjugate | $g^*(t)$ | $G^*(-f)$ |
| 6 | Temporal derivative | $\frac{d^n}{dt^n}g(t)$ | $(j2\pi f)^n.G(f)$ |
| 7 | Spectral derivative | $(-j2\pi t)^n.g(t)$ | $\frac{d^n}{df^n}G(f)$ |
| 8 | Reciprocity | $G(t)$ | $g(-f)$ |
| 9 | Linearity | $A.g(t) + B.h(t)$ | $A.G(f) + B.H(f)$ |
| 10 | Multiplication | $g(t).h(t)$ | $G(f) * H(f)$ |
| 11 | Convolution | $g(t) * h(t)$ | $G(f).H(f)$ |
| 12 | Delta function | $\delta(t)$ | $1$ |
| 13 | Constant | $1$ | $\delta(f)$ |
| 14 | Rectangular function | $\mathbf{rect}\{t\} \triangleq \begin{cases} 1 & \text{if } \lvert t\rvert < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$ | $\mathbf{sinc}\{f\} \triangleq \frac{\sin(\pi f)}{\pi f}$ |
| 15 | Sinc function | $\mathbf{sinc}(t)$ | $\mathbf{rect}\{f\}$ |
| 16 | Unit step function | $u(t) \triangleq \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$ | $\frac{1}{2}\delta(f) - \frac{j}{2\pi f}$ |
| 17 | Signum function | $\mathrm{sgn}(t) \triangleq \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$ | $-\frac{j}{\pi f}$ |
| 18 | decaying exp (two-sided) | $e^{-\lvert t\rvert}$ | $\frac{2}{1+(2\pi f)^2}$ |
| 19 | decaying exp (one-sided) | $e^{-\lvert t\rvert}.u(t)$ | $\frac{1-j2\pi f}{1+(2\pi f)^2}$ |
| 20 | Gaussian function | $e^{-\pi t^2}$ | $e^{-\pi f^2}$ |
| 21 | Lambda function | $\Lambda\{t\} \triangleq \begin{cases} 1-t & \text{if } 0 \le t \le 1 \\ 1+t & \text{if } -1 \le t \le 0 \end{cases}$ | $\mathbf{sinc}^2\{f\}$ |
| 22 | Repeated function | $\mathbf{rep}_T\{g(t)\} = g(t) * \mathbf{rep}_T\{\delta(t)\}$ | $\left\lvert\frac{1}{T}\right\rvert \mathbf{comb}_{\frac{1}{T}}\{G(f)\}$ |
| 23 | Sampled function | $\mathbf{comb}_T\{g(t)\} = g(t).\mathbf{rep}_T\{\delta(t)\}$ | $\left\lvert\frac{1}{T}\right\rvert \mathbf{rep}_{\frac{1}{T}}\{G(f)\}$ |

The graph below shows the Tail function $\mathbf{T}\{x\}$ which represents the area from $x$ to $\infty$ of the Gaussian probability density function N(0,1), i.e.

$$\mathbf{T}\{x\} = \int_x^\infty \frac{1}{\sqrt{2\pi}} . \exp\left(-\frac{y^2}{2}\right) dy$$




Note that if $x > 6.5$ then $\mathbf{T}\{x\}$ may be approximated by $\mathbf{T}\{x\} \approx \frac{1}{\sqrt{2\pi}.x} . \exp\left\{-\frac{x^2}{2}\right\}$